

LECTURE 32: MATH1061/7861
TOPICS: INTRODUCTION TO PROBABILITY
DATE: WEDNESDAY MAY 13, 2026

Learning objectives:

- (1) Use counting techniques to compute probabilities.
- (2) Reinforce techniques to solve counting problems.

Q1: There are ~~30~~⁴¹ people in our class. What is the probability that at least two people share a birthdate?

Easier to do the complement event:

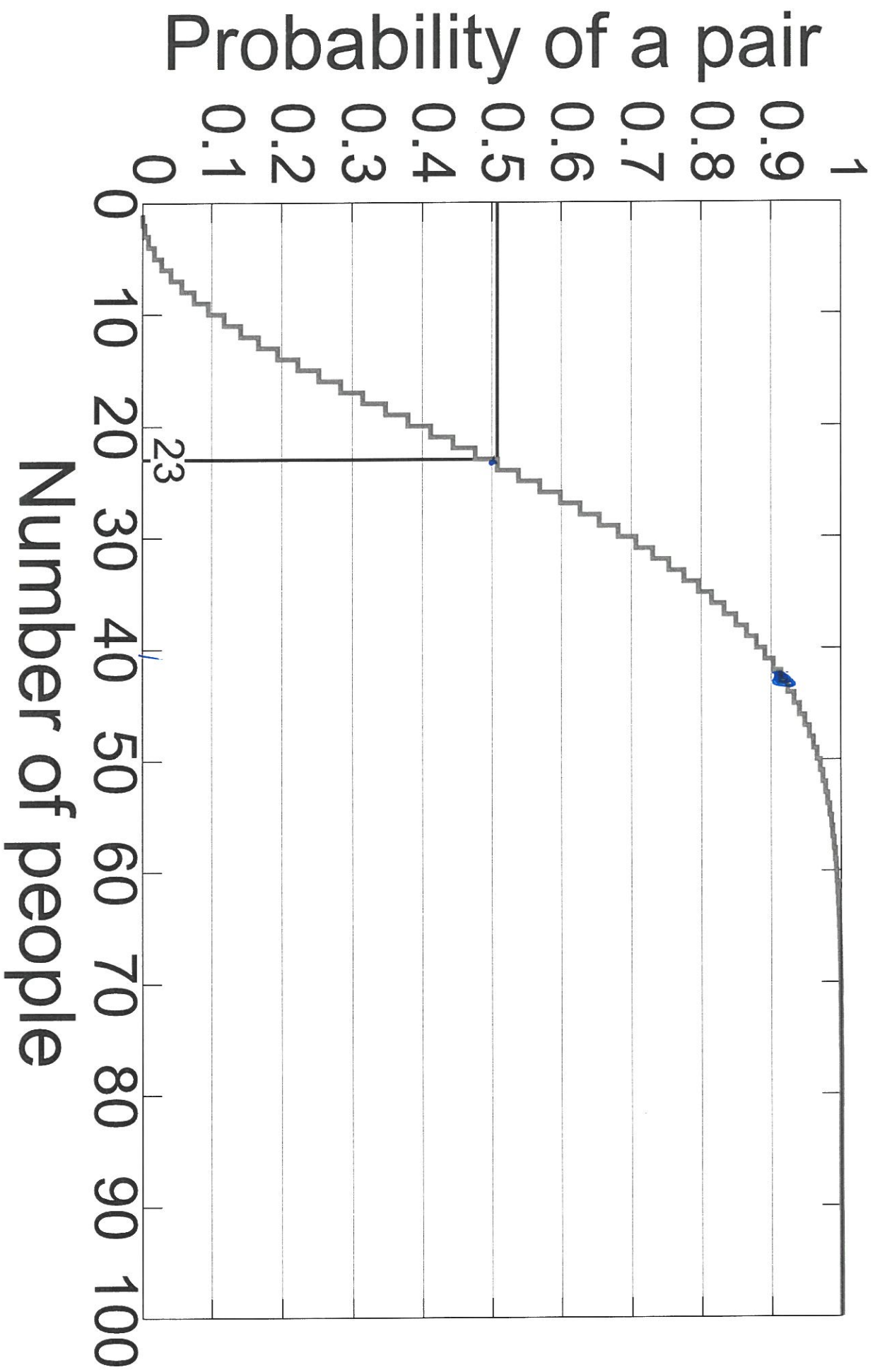
Probability that no two people share a birthday equals $= \frac{365 \times 364 \times \dots \times 324}{365^{41}}$

$$= \frac{365}{365} \times \frac{364}{365} \times \frac{363}{365} \times \dots \times \frac{324}{365}$$

$$= 0.0968$$

$\Rightarrow$ probability that no two people share a birthday $\approx 9.68\%$

$\Rightarrow$ Probability that at least two people share a birthday $= 90.32\%$



Month	Relative Frequency of Births
January	8.0%
February	7.4%
March	8.5%
April	8.3%
May	8.5%
June	8.3%
July	8.9%
August	9.1%
September	9.2%
October	8.7%
November	8.1%
December	7.0%

Q2: A 5-card hand is drawn at random from a standard 52-card deck. What is the probability that the hand contains exactly 2 aces?

$$\text{Total number of hands} = \binom{52}{5}$$

$$\begin{aligned}\text{Number of hands with 2 aces} &= \frac{4 \times 3}{2} \times \binom{48}{3} \\ &= \binom{4}{2} \times \binom{48}{3}.\end{aligned}$$

$$\text{Probability} = \frac{\binom{4}{2} \binom{48}{3}}{\binom{52}{5}} = 0.0399 \approx 4\%.$$

Exercise: Compute probability for all poker hands.

Q3: If a committee of 5 is chosen at random from a group of 25 mathematicians and 25 physicists, what is the probability that at least one mathematician will be chosen?

$$\begin{aligned} \text{Total number of possible committees} \\ = \binom{50}{5} \end{aligned}$$

$$\begin{aligned} \# \text{ of committees with no mathematicians} \\ = \binom{25}{5}. \end{aligned}$$

$$\begin{aligned} \text{Probability of having no mathematician} & \frac{\binom{25}{5}}{\binom{50}{5}} \\ & \approx 2.5\%. \end{aligned}$$

$$\begin{aligned} \text{Probability of having at least} \\ \text{one math. is} & = 1 - \frac{\binom{25}{5}}{\binom{50}{5}} \\ & \approx 97.5\% \end{aligned}$$